EAS5450 Quiz: 8:35 – 8:45AM
Canvas Quizzes

We see a world in which every person has access to actionable health information at the time it matters. A world in which no one ever has to say, "if only I'd known sooner."

- Elizabeth Holmes, 2014

EAS5450: Class #27

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The Edison – FDA Class II Medical Device

- CMS oversaw clinical laboratory quality standards
- Detecting up to 240 diseases with just a finger prick of blood
- Low cost, accessible, efficient alternative to traditional blood tests

Nanotainer

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Elizabeth Holmes

- 2003 dropped out of Stanford
- Founded Theranos
- Raised $9.6M VC by 2006
- 2015 Forbes Most Powerful Women

Ramesh Balwani

- 2009 President & CEO
- Invested $13M interest-free
- No medical device or life science training

- Prominent men on Board of Directors
- High-profile investors & VCs

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Framework for this lecture

- Assume that those committing fraud or behaving unethically are not inherently bad or deviant (even though some may be)
- Assume they did not start their business or endeavor with the intention of committing fraud or unethical behavior (even though some may have)
- With these assumptions in mind….how does it happen??

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Understanding Cognitive Bias

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Recall…

Cognitive Biases

Anchoring on Misleading Perception Impairs Rational Decision-Making

Prof. Daniel Kahneman
Nobel Prize winner

- **Optimism**
  - Reality distortion
- **Planning Fallacy**
  - Tendency to overestimate benefits
  - Tendency to underestimate costs, durations, risks
- **Confirmation Bias**
  - Over-weighting validating results
  - Ignoring contrary results
- **Sunk Costs**
  - "Wasteful" to discard past effort, cost

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Cognitive bias can also contribute to ethical failures

Cognitive Bias[1]	Description	Theranos case????
Self-Interest Bias	Self interest over other factors	
Affect Heuristic	Emotion (love of idea) drives decision	
Groupthink	Discourages individual responsibility	
Saliency Bias	Over-reliance on analogous situation	
Confirmation Bias	Favoring facts that support your original thesis	
Availability Bias	Over-reliance on readily available info	
Anchoring Bias	Over-reliance on early info	
Halo Effect	Assumption that a talented person/team can't fail	
Sunk-Cost Fallacy	Protecting against "wasting" past efforts	
Overconfidence Bias	More confident than objectively warranted	
Disaster Neglect	Underestimate worst case	
Loss Aversion	Loss is more feared than gain is desired	

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[1]List from: *Before You Make That Big Decision*. Khaneman et. Al. Harvard Business Review. 2011

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Awareness of common biases can *improve* decision making

Before You Make That Big Decision
by Daniel Khaneman et. al. (the Godfather of Behavioral Economics!)

Ask Yourself
- **Self-Interest Bias**
- **Affect Heuristic**
- **Groupthink**

Ask the Recommenders
- **Saliency Bias**
- **Overconfidence Bias**
- **Confirmation Bias**
- **Availability Bias**
- **Anchoring Bias**
- **Halo Effect**

Ask About the Proposal
- **Sunk-Cost Fallacy**
- **Disaster Neglect**
- **Loss Aversion**

It's not just about staying out of trouble - it improves performance!

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We all enjoy a good story

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2 Modes of Continuously Active Thinking

- **Intuitive Thinking**
  - Flows effortlessly without conscious focus
  - Determines thoughts and actions most of the time
  - We just do things
  - Suspended disbelief
- **Reflective Thinking**
  - Effortful, deliberate, rule-based
  - Requires an act of will
  - Disbelief not suspended
- **When Hearing Stories**
  - Intuitive thinking digests the story & develops a narrative
  - In the absence of conscious focus, suspends disbelief
  - Suppresses alternative stories
  - With no thought of cognitive biases

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In a world where storytelling rules, we must be even more mindful of ethics… and be careful to not be duped!

- Storytelling is a very effective way to convey your vision and ideas
  - It has been embraced in the business world
  - It has been embraced in business education (including in EAS5460 & EAS5410)
  - Stories are told mainly in the teller's interests
- But an emotional story is also the tool of cheats

- We want to believe
  - We don't *actively* suspend disbelief…our disbelief is automatically suspended
  - Not suspending disbelief requires an act of will not the other way around

Therefore:
- To employ storytelling, we must have a rock-solid foundation of honesty and ethics
- We must be aware of storytelling's power to avoid being duped ourselves (trust but verify)

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Four Principles of Biomedical Ethics

- **Respect for autonomy**
  - Tell the truth.
  - Respect the privacy of others.
  - Protect confidential information.
  - Obtain consent for interventions with patients.
  - When asked, help others make important decisions.

- **Nonmaleficence (intentionally, recklessly, unintentionally, carelessly)**
  - Do not kill, cause pain or suffering, or incapacitate.
  - Do not cause offense.
  - Do not deprive others of the goods of life.

T.Beauchamp & J.Childress, *Principles of Biomedical Ethics*, Oxford University Press, 2001

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Four Principles of Biomedical Ethics (cont'd)

- **Beneficence**
  - Protect and defend the rights of others.
  - Prevent harm from occurring to others.
  - Remove conditions that will cause harm to others.
  - Help persons with disabilities.
  - Rescue persons in danger.
- **Justice**
  - To each person an equal share.
  - To each person according to need.
  - To each person according to effort.
  - To each person according to contribution.
  - To each person according to merit.
  - To each person according to free-market exchanges.

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Elizabeth Holmes, Storyteller

I grew up with these stories of greatness … and about people deciding not to spend their lives on something purposeful, and what happens to them when they make that choice – the impact on character and quality of life.

- Elizabeth Holmes

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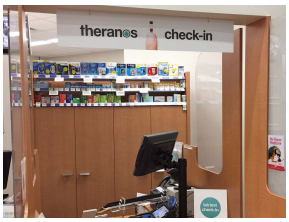
Duped…

We can't not pursue this. We can't risk a scenario where CVS has a deal with them in six months and it ends up being real.

- R. Van den Hoof, Walmart executive
Prior to $150M Walton family investment



… who else?

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Beware the slippery slope - "Just this once" never ends well

"I started to go off the tracks, and I was able to convince myself that this was, you know, a temporary situation"

- Bernie Madoff
Interview from prison 2011

Remember the words of our guest Ted Schlein:
"Integrity is binary"

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Use radical transparency with investors and employees to *improve* performance

Triple A Rubric

Part of corporate maturation process

Builds trust with customers, aids recruitment, increases investor comfort

Acts as a forcing function for *internal* accountability, reliability, control

Carries over to improve internal functions such as reporting, financial forecasting, etc.

By aligning, puts employees and investors in position to provide help

It's not just about staying out of trouble - it improves performance!

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Theranos and Culture

"Culture eats strategy for lunch"
- M. Zisman 12/1/22

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Culture is an Effective Leadership Tool if…

- **It is strategically relevant:**
  - Group norms shape behavior more powerfully than either monetary rewards or physical environment.
  - Best achieved in young firms.
- **The culture promotes innovation and change:**
  - More value on creative skills than on critical skills.
  - Encourages inter-personal risk-taking, expressing ideas.
  - People feel empowered.
  - Leaders move quickly to implement promising ideas and change.
- **Employees are trained and socialized in the culture.**
- **Leaders consistently "walk the talk" …**
  - Employees attend vigilantly to leaders' behavior.

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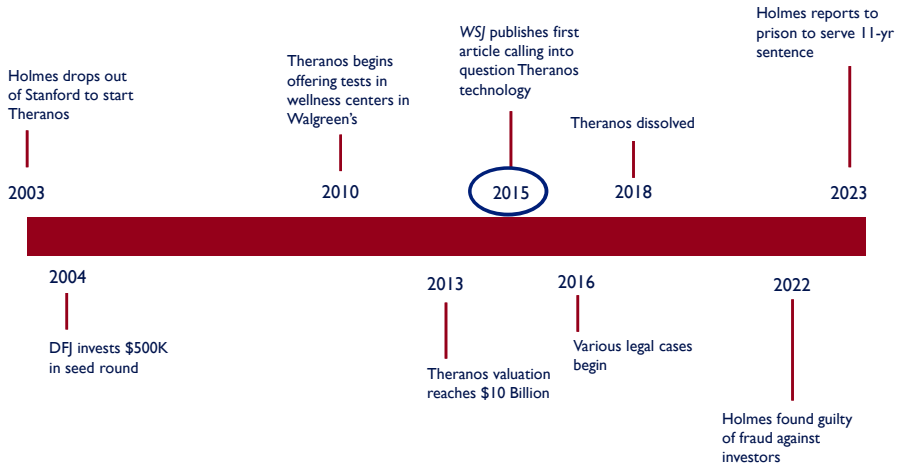
Theranos timeline
Total capital raise = ~$750 million



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Change in styles …

2015 Prize Winner

2023 + next 10 years

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Questions…

1. What were the underlying conditions that provided fertile ground upon which Elizabeth Holmes could commit fraud?
2. Was there any point during Theranos's operational life where a change in strategy may have saved the company?
3. How did it go on so long? Why didn't anyone stop her sooner?

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Will we ever be able get all routine bloodwork from a single drop of blood in 20 minutes? Maybe?!

truvian™

Data before hype
- Head-to-head results vs. central lab machine presented at respected medical conference

Radical Transparency
- Encourage investors to get test done on Truvian machine and compare to central lab results of their choosing
- Machine has transparent casing and investors can operate themselves!

Setting Realistic expectations
- Today – 300uL sample in 25-30 minutes
- Aspiration – smaller and faster

Attempting to get FDA approval in 2025

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Disclaimer: Domain and MEC backed (Prof. Halak has a financial interest!)

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